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Fast and accurate modelling of twin-screw compressors: A generalised low-order approach

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ABSTRACT

Twin-screw compressors (TSC) are commonly used in heat pump processes due to their robustness and flexibility. They exhibit two core properties, i.e. the swept volume and the built-in volume ratio (BVR), which heavily influence their capacity limits and off-design efficiency. This work presents a new low-order model (i.e. a polynomial model), which can accurately predict a TSC's behaviour. The model uses the external pressure ratio and volumetric compressor inlet flow rate to calculate isentropic efficiency and compressor speed. The input parameters are normalised with a reference flow rate (calculated from the swept volume) and the BVR, respectively. This results in a generalised model of low numerical cost, which can be used for explorative studies independent of the specific machine size and BVR. A gain in computational speed by a factor of 375 is achieved compared to a semi-empiric reference model. The model displays very good predictive accuracy when used to predict the performance of machines with similar BVRs, but different sizes. A mean deviation from the manufacturer data of 4.29 %, 0.88 °C and 1.38 % for the shaft power, the outlet temperature and the compressor speed can be observed, respectively. When there is a difference in size and BVR, the prediction accuracy is still reasonable but significantly declines for small and very large pressure ratios. Nevertheless, the proposed new approach extends the state-of-the-art by introducing a low-order model, which combines the advantages of low computational cost, high accuracy, physically correct predictions over a wide operational range and scalability to different machine capacities and BVRs. The validation for different fluids indicates a good general prediction accuracy relatively independent of the used fluid.

1. Introduction

1.1. TSC modelling in the literature

With an increasing demand for efficient and sustainable heating and cooling, vapour compression heat pumps have become the cornerstone of many industrial and residential chilling and heating applications. The key component of these heat pumps is often a volumetric compressor such as a lubricated twin-screw compressor (TSC) or scroll compressor, which are also used in other applications such as air and process gas compression. These machines are widely acknowledged for their robustness and flexibility [1]. Besides their use in heat pumps, twin-screw machines are of high relevance in various application fields, such as Organic Rankine Cycles (ORCs), where they are used as expanders [2]. Their high robustness against droplet formation also makes them attractive for two-phase expansion processes, as described in more detail by Öhman [3] and Dawo et al. [4], as well as for reversible heat pump/ORC systems [5]. Thus, the design, evaluation and control of facilities using TSCs require component models that can accurately predict the machine performance during full- and part-load operation. Hence, different modelling approaches for volumetric compressors have been proposed in literature, which can roughly be divided into empiric, semi-empiric and geometric models [6]. In this context, models originally developed for scroll machines are often adapted for TSCs due to the shared basic working principles of both machines. The accuracy and extrapolation capability but also the computational cost increases from empiric over semi-empiric to geometric models. For applications that require low computational cost, empiric and semi-empiric models are favourable.

Empiric models are usually based on a set of polynomials that are fitted to the operational data of a specific machine, requiring no knowledge of the machine geometry and working principle. Their advantage

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of low computational cost is opposed by a very limited extrapolation capability [6]. In addition, operational ranges not covered by the fitting data can result in significant inaccuracies [7]. Browne and Bansal (2002) present an empiric model for twin-screw compressors used for transient simulations of vapour-compression packaged liquid chillers, determining the isentropic efficiency from a biquadratic function of refrigerant flow rate and system pressure [8]. Their model shows good prediction accuracy within $\pm 10\,\%$ of the experimental reference values. Liu et al. (2012) present an empiric model for twin-screw refrigeration compressors showing very high accuracy when compared to experimental data but relying on many fitted parameters [1]. Moreover, Navarro-Peris et al. (2013) present an interesting approach to correlate the non-dimensional mass flow rate and power input of different volumetric compressors to only five empirically determined parameters [9]. However, their approach is based on a previous model for piston compressors, indicating limited validity for TSCs. Markham and Cutbirth (2016) propose an empiric polynomial model for an asymmetric TSC, yielding very good agreement with experimental data [10]. Their results should be considered carefully, however, as the authors do not base their fitting procedure on an independent fitting and validation data set. Finally, Kosmadakis and Neofytou (2022) propose an empiric model for a small-scale TSC used in annual simulations of a reversible Organic Rankine Cycle [11]. Their model yields the volumetric and isentropic compressor efficiency based on the applied pressure or volume ratio and the volumetric suction flow rate.

Semi-empiric models employ physics equations to describe the thermodynamic and mechanical processes inside the machine as well as empiric parameter equations, which are fitted to operational data. This requires more extensive knowledge on the specific machine and often needs iterative solvers leading to higher computational costs. Giuffrida (2016) proposed a simple semi-empiric model for TSCs with good accuracy based on six experimentally determined parameters [12]. Another semi-empiric model for TSCs developed by Zhao et al. (2016) includes several different leakage paths inside the compressor and mostly shows model deviations smaller than $5\,\%$ for volumetric and isentropic efficiency as well as outlet temperature [13]. Furthermore, Salts et al. (2019) present a detailed semi-empiric modelling framework for oil-injected TSCs, accounting for a large variety of thermo-physical processes inside a TSC [14].

Higher-order TSC models, also referred to as geometric or deterministic models, can be divided in mathematical chamber models and computational fluid dynamics (CFD) approaches. They show the best accuracy and are most reliable at predicting part-load behaviour. However, they usually show significantly higher computational times than the other two approaches and also require extensive knowledge about constructional details of the observed machines, which is not always available [7]. Hanjalić and Stošić (1997) describe a numerical model able to precisely describe the thermodynamic and fluid dynamic processes inside the compressor, which can be used to optimise rotor profiles [15]. Lee et al. (2001) present a mathematical chamber model able to predict the relevant process variables of a twin-screw air compressor with satisfying accuracy and draw conclusions on the effect of tip speed and oil injection angle on leakage and efficiency [16]. Seshaiah et al. (2007) developed an elaborate and accurate mathematical model that puts a special focus on the effect of oil and internal heat transfer in oil-injected TSCs [17]. Buckney et al. (2013) validate a geometry model for two different rotor profiles, which is able to include the effect of rotor temperatures on the internal leakage [18]. Chamoun et al. (2013) make use of a *Modelica*-based chamber model to predict the dynamic behaviour of a water-injected TSC with biphasic flow [19] . Wu and Tran (2016) present a dynamic multi-body model for TSCs with a focus on the forces imposed on the rotors and bearings during the compression cycle and load changes [20]. Kennedy et al. (2017) compare a 3D CFD model to a 1D chamber model and conclude better prediction accuracy for the chamber model when compared to test data [21]. Andres et al. (2018) present a CFD model for a two-stage intercooled TSC solved on a single computational domain able to predict the compressor behaviour with acceptable accuracy compared to experimental results [22]. Basha et al. (2019) compare different approaches to describe the oil-fluid-machine interaction in a CFD model of a TSC [23]. Due to their inherent complexity, TSCs are mostly described with deterministic approaches in literature, and the number of publications on deterministic TSC models is much higher compared to semi-empiric and empiric models. Since this work is more concerned with (semi-)empirical models, only a few deterministic approaches were presented.

For a more exhaustive overview on deterministic models the interested reader is referred to Chamoun et al. (2013) [19], Byeon et al. (2014) [24], Wu et al. (2016) [20], Kovacevic et al. (2014) [25] or Andres et al. (2018) [22].

1.2. Motivation and contribution

Two constructional parameters, i.e. the swept volume and the built-in volume ratio (BVR), heavily influence the capacity limits and part-load efficiency of TSCs. At the same time, these parameters are used in most (semi-)empiric models to describe the machine performance making it impossible to use the models for any other machine than the one the model was fitted on. This work presents a new polynomial model that accurately predicts the behaviour of a TSC with a broad scope of validity. In the first step, a highly accurate semi-empiric model is used to generate performance data for a large operational field. Then a polynomial model with linearisation for high pressure ratios is fitted to this data. The model uses the external pressure ratio and volumetric compressor inlet flow rate to calculate isentropic efficiency and compressor speed. Both input parameters are normalised with a reference flow rate and the BVR, respectively. This results in a generalised model of low numerical cost, which can be used for explorative studies independent of the specific machine size and BVR. The novelty of the model is defined by two key features, that separate it from previously proposed models. On one hand, the linearisation enables the model to give physically meaningful predictions outside of the operational range used in the fitting procedure, which is not the case for most published polynomial models. On the other hand, it accurately predicts part-load efficiency for different machine capacities, which is also not possible with most published low-order models. Potential applications of the polynomial model include the optimal sizing of heat pump systems considering realistic operational boundaries and the scale-up of lab-scale and pilot facilities to commercial sizes. Finally, the model can easily be adapted for twin-screw expanders, which are of high relevance for various applications such as Organic Rakine Cycles.

2. Model development

2.1. Base model and data generation

Due to limitations in available operational data, the performance data that is used to fit and, in parts, validate the polynomial model is generated from a semi-empiric TSC model. The semi-empiric model used in this work is a modified version of the one proposed by Giuffrida (2016) [12]. This section first gives an explanation of the model and the operational data it is based on. Afterwards, the data set generation for fitting the polynomial model is described. The model implementation and solving are conducted in *MATLAB 2021a* [26] with the fluid properties being calculated from *REFPROP 10* [27].

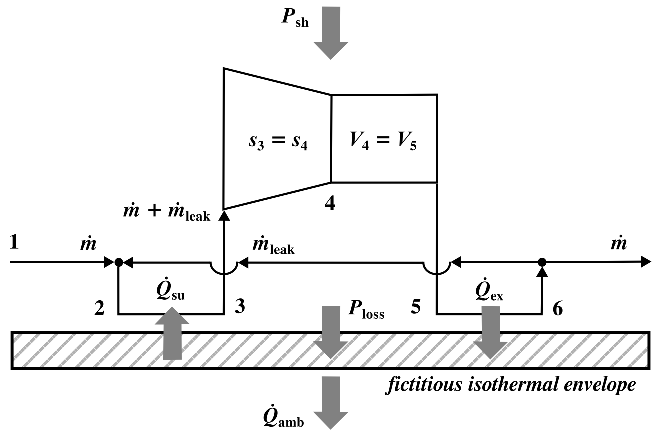


Fig. 1. Semi-empiric model structure proposed by Giuffrida (2016) [12].

2.1.1. Semi-empiric model

The semi-empiric base model, which was adapted for data generation, is depicted in Fig. 1 and considers the most relevant characteristics of a TSC. This includes mechanical losses in the compressor (P_{loss}) as well as internal leakage ($\dot{m}_{\mathrm{leak}}$), suction heat transfer between the incoming working fluid and the compressor housing ($\dot{Q}_{\mathrm{su}}$) and heat losses to the surrounding ($\dot{Q}_{\mathrm{amb}}$). To account for the effect of over- and under-compression related to the BVR, the base model applies an isentropic compression (3→4) followed by an isochoric compression (4→5). In contrast to the original model, heat transfer between the pressurised gas and the housing ($\dot{Q}_{\mathrm{ex}}$) is neglected in this work. This assumption was validated in pre-studies, which also accounted for the respective heat flow but returned very small values for the exhaust side heat transfer coefficient. The construction and the precise internal flow paths of the compressor (and the according heat transfer area in the exhaust line) are mostly proprietary knowledge of the manufacturer and only available to a limited extent. Hence, no definitive explanation can be given for the small heat transfer coefficient. However, several arguments can be made to support this observation. Firstly, the original model publication also indicates significantly smaller exhaust side heat transfer coefficients [12]. Secondly, the assumption of a fictitious isothermal envelope (or compressor housing) could be an explanation. Older documentation of the compressors observed in this work indicates a larger number of bearings on the exhaust side of the compressor [28], leading to higher local friction heat production and higher local housing temperatures. The experimental exhaust side temperature gradient between pressurised gas and housing is thus lower and results in a low heat transfer at the exhaust side. Using a homogeneous housing temperature, the semi-empirical model naturally produces a higher temperature gradient at the exhaust side. Consequently, the fitting procedure reduces the exhaust side heat transfer coefficient to match the low heat transfer in the exhaust line from the experimental data.

In the first step, the leakage mass flow rate $\dot{m}_{\mathrm{leak}}$ is determined by assuming a single leakage path connecting the exhaust and inlet stream. The leakage path is modelled as an isentropic flow through a convergent nozzle with the cross-sectional area A_{leak}. The resulting mass flow rate can be determined by

$$\dot{m}_{\mathrm{leak}} = \frac{A_{\mathrm{leak}}}{v_{\mathrm{leak}}} \cdot \sqrt{2 \cdot (h_6 - h_{\mathrm{leak}})} \ . \tag{1}$$

The specific enthalpy h_{leak} and specific volume v_{leak} at the throat of the nozzle are calculated based on the specific entropy of the exhaust gas s_6 and the pressure at the throat. The latter is the maximum between the suction pressure p_1 and the critical pressure in the nozzle $p_{\mathrm{crit,leak}}$ which is defined as

$$p_{\mathrm{crit,leak}} = p_6 \cdot \left(\frac{2}{\gamma+1}\right)^{\dfrac{\gamma}{\gamma-1}} \tag{2}$$

with γ being the isentropic exponent. For further background on the calculation of $p_{\mathrm{crit,leak}}$ the reader is referred to Giuffrida [12] or Lemort et al. [29]. Eq. (1) allows the determination of the swept mass flow rate $\dot{m}_{\mathrm{s}}$ by combining the suction mass flow rate $\dot{m}$ and the leakage mass flow rate:

$$\dot{m}_{\mathrm{s}} = \dot{m} + \dot{m}_{\mathrm{leak}} \tag{3}$$

The adiabatic mixing of both flows is followed by heat exchange with the compressor housing, which is modelled as a fictitious isothermal envelope of the temperature T_{w}. The heat flow to the fluid $\dot{Q}_{\mathrm{su}}$ (with the index su referring to the suction side) can be described in dependence of the fluid's isobaric heat capacity c_{p} by the equation

$$\dot{Q}_{\mathrm{su}} = \dot{m}_{\mathrm{s}} \cdot c_{\mathrm{p},2} \cdot (T_{\mathrm{w}} - T_2) \cdot \left[1 - e^{\left(\dfrac{-UA_{\mathrm{su}}}{\dot{m}_{\mathrm{s}} \cdot c_{\mathrm{p},2}}\right)}\right] . \tag{4}$$

The flow-regime dependency of the lumped heat transfer factor UA_{su} is accounted for by assessing the relation between $\dot{m}_{\mathrm{s}}$ and a nominal mass flow rate $\dot{m}_{\mathrm{nom}}$ for which the model parameter $UA_{\mathrm{su,nom}}$ is fitted:

$$UA_{\mathrm{su}} = UA_{\mathrm{su,nom}} \cdot \left(\frac{\dot{m}_{\mathrm{s}}}{\dot{m}_{\mathrm{nom}}}\right)^{0.8} \tag{5}$$

This equation is based on the *Dittus–Boelter*-correlation [30] for fully developed flows in smooth circular pipes defined as:

$$Nu = 0.023 \cdot Re^{0.8} \cdot Pr^{0.4} \tag{6}$$

Although the geometry and flow conditions in the compressor in- and outlet most likely do not comply with the requirements for the *Dittus–Boelter*-correlation, the approach has proven to be sufficiently reliable in previous publications [12,29].

Since the in- and outlet geometry does not change and the temperature-dependency of fluid properties are neglected for Eq. (6), the Prandtl-number Pr can be neglected and the Nusselt-number Nu (and hence the heat transfer parameter UA) is solely dependent from the Reynolds-number Re. Due to the unchanging geometry and the simplifying assumption of constant fluid properties, the latter can be replaced by the mass flow rate leading to Eq. (5).

Knowing the fluid state after suction warming (i.e the specific volume v_3), the compressor speed n can be calculated with the swept compressor volume V_{s}:

$$n = \frac{v_3 \cdot \dot{m}_{\mathrm{s}}}{V_{\mathrm{s}}} \tag{7}$$

The total internal power P_{int} required for both continuous compression steps is described by

$$P_{\mathrm{int}} = \dot{m}_{\mathrm{s}} \cdot \left((h_4 - h_3) + v_4 \cdot (p_5 - p_4)\right) \ . \tag{8}$$

The state variables after the isentropic compression (i.e. the specific enthalpy h_4 and the pressure p_4) are calculated from the specific inlet entropy s_3 and the specific volume v_4, which is in turn calculated from the inlet state by the equation

$$v_4 = \frac{v_3}{\mathrm{BVR}} \ . \tag{9}$$

The first contributor to the mechanical losses $P_{\mathrm{loss,1}}$ is modelled as a constant fraction of the internal power using the dimensionless model parameter $a_{\mathrm{tl,1}}$:

$$P_{\mathrm{loss,1}} = a_{\mathrm{tl,1}} \cdot P_{\mathrm{int}} \tag{10}$$

A second mechanical loss $P_{\mathrm{loss,2}}$, which is depending on the compressor speed, can be calculated by the equation

$$P_{\mathrm{loss,2}} = a_{\mathrm{tl,2}} \cdot V_{\mathrm{s}} \cdot \mu \cdot \left(\frac{\pi \cdot n}{30}\right)^2 \ , \tag{11}$$

including the dynamic viscosity of the lubrication oil μ as well as the dimensionless model parameter $a_{\mathrm{tl},2}$. Using the Eqs. (8), (10) and (11) allows to calculate the total required shaft power P_{sh} as

$$P_{\mathrm{sh}} = P_{\mathrm{int}} + P_{\mathrm{loss},1} + P_{\mathrm{loss},2} \ . \tag{12}$$

In order to calculate a heat balance around the compressor housing, also the heat loss to the surrounding $\dot{Q}_{\mathrm{amb}}$ has to be considered:

$$\dot{Q}_{\mathrm{amb}} = b_{\mathrm{hl}} \cdot (T_{\mathrm{w}} - T_{\mathrm{amb}})^{1.25} \tag{13}$$

It depends on the ambient temperature T_{amb} and the heat transfer factor to the surrounding b_{hl}. The exponent in Eq. (13) is due to its derivation from natural convection phenomena. For more information please refer to Giuffrida [12].

For the model to be solved, the energy balances around the compressor housing and around the whole compressor have to be satisfied. The first is defined as

$$P_{loss,1} + P_{loss,2} - \dot{Q}_{amb} - \dot{Q}_{su} = 0 \ , \tag{14}$$

while the latter can be written as

$$P_{sh} - \dot{Q}_{amb} + \dot{m} \cdot (h_1 - h_6) = 0 \ . \tag{15}$$

It is worth mentioning, that the mechanical losses are completely transferred to the compressor housing as friction heat. Since the thermal envelope temperature and the outlet state are not known a priori, they have to be determined iteratively by numerical minimisation of the residuals from Eqs. (14) and (15).

According to the previous section, the base model is fully defined by the seven parameters A_{leak} in mm^2, the nominal suction heat transfer coefficient $UA_{\mathrm{su,nom}}$ in W/K, the external heat transfer coefficient b_{hl} in W·K$^{-5/4}$, the dimensionless torque loss coefficients $a_{\mathrm{tl},1}$ and $a_{\mathrm{tl},2}$, the machine's swept volume V_{s} in L and finally its BVR. The swept volume V_s and the BVR are constant parameters defined by construction and hence do not need to be identified in the fitting procedure. The results of the fitting procedure together with the determined model parameters and the model validation results are presented in the results Section 3.1.1.

2.1.2. Data set generation

The model described in the previous section is used to generate a set of performance data distributed over a wide operational range with respect to flow rates (translating into compressor speeds) and pressure ratios. To determine the operational limits regarding the flow rate, a minimum and maximum speed of 500 rpm and 4500 rpm, respectively, are assumed, which is roughly the machine's speed range proposed by the manufacturer. In this context, the volumetric suction flow rate can be determined as the product of compressor speed and the compressor's swept volume, neglecting internal leakages for simplicity reasons. In order to account for different inlet conditions, inlet pressures of 1, 3 and 5 bar are considered and the inlet temperature is calculated assuming a superheating of 10 °C for the suction stream. The inlet mass flow rate is calculated based on the volumetric suction flow rate and the inlet density at the respective suction temperature and pressure. A range between 1.5 and 8 is used for the applied external pressure ratio, and the outlet pressure is obtained by multiplying the inlet pressure and external pressure ratio. The external pressure ratio is limited to a range between 1.5 and 5 for the suction pressure of 5 bar to avoid unreasonably high outlet pressures. With the boundary values for flow rate and pressure ratio at hand, a mesh grid of 400 (20 x 20) evenly distributed operational points is created for each suction pressure. This yields a data set of 1200 operational data points for the semi-empiric model, representing a wide variety of operating conditions for the compressor. The resulting data set and the observable behaviour are presented and discussed in the results Section 3.1.2.

2.2. Polynomial model

In the next step, the data generated as described in the previous section is used to develop a generalised low-order model based on polynomials.

2.2.1. Model equations

A normalisation of the volumetric suction flow rate $\dot{V}_{\mathrm{suc}}$ and the applied external pressure ratio r_{p} with a reference flow rate and the machine's BVR aims at providing a generalised model capable to predict realistic machine performance independent from the specific machine size and BVR. The normalised suction flow rate $\dot{V}_{\mathrm{suc,N}}$ and external pressure ratio $r_{\mathrm{p,N}}$ are defined as

$$\dot{V}_{\mathrm{suc,N}} = \frac{\dot{V}_{\mathrm{suc}}}{V_{\mathrm{s}} \cdot n_{\mathrm{ref}}} \qquad \text{and} \tag{16}$$

$$r_{\mathrm{p,N}} = \frac{r_{\mathrm{p}}}{\mathrm{BVR}} \ . \tag{17}$$

Based on the usual design speed of refrigeration compressors, the reference compressor speed n_{ref} is set to 50 Hz (equal to 3000 rpm) in this work. In order to account for over- and under-compression losses, the applied volume ratio and its relation to the machine's BVR are of major importance. However, only the applied pressure ratio is known a priori, while the specific volume at the compressor outlet depends on the outlet temperature and is hence depending on the initially unknown machine efficiency. This usually requires iterative solving strategies, which should be avoided for low-order models. Consequently, a first polynomial correlating the ratio ($r_{\mathrm{v,p}}$) of the applied volume ratio r_{v} to the applied pressure ratio r_{p} is fitted to the generated data. A polynomial of the general form

$$\begin{aligned} r_{\mathrm{v,p}} = {} & c_1 + c_2 x + c_3 y + c_4 x^2 + c_5 xy + c_6 y^2 + c_7 x^3 + c_8 x^2 y + c_9 x y^2 \\ & + c_{10} y^3 + c_{11} x^4 + c_{12} x^3 y + c_{13} x^2 y^2 + c_{14} x y^3 + c_{15} x^5 + c_{16} x^4 y \\ & + c_{17} x^3 y^2 + c_{18} x^2 y^3 \end{aligned} \tag{18}$$

is used, where x and y need to be substituted by $\dot{V}_{\mathrm{suc,N}}$ and $r_{\mathrm{p,N}}$, respectively. The applied volume ratio r_{v} is then obtained from

$$r_{\mathrm{v}} = r_{\mathrm{v,p}} \cdot r_{\mathrm{p}} \ . \tag{19}$$

The obtained polynomial coefficients c_i of the fitting procedure are summarised for all relevant polynomials in Table 2.

A similar approach is taken to calculate the volumetric efficiency. Here, the polynomial

$$\begin{aligned} \eta_{\mathrm{vol}} = {} & c_1 + c_2 x + c_3 y + c_4 x^2 + c_5 xy + c_6 y^2 + c_7 x^3 + c_8 x^2 y + c_9 x y^2 \\ & + c_{10} y^3 + c_{11} x^4 + c_{12} x^3 y + c_{13} x^2 y^2 + c_{14} x y^3 + c_{15} y^4 + c_{16} x^5 \\ & + c_{17} x^4 y + c_{18} x^3 y^2 + c_{19} x^2 y^3 + c_{20} x y^4 \end{aligned} \tag{20}$$

using $\dot{V}_{\mathrm{suc,N}}$ and r_{p} for x and y is fitted to the generated dataset. It returns a value between zero and one. Since the driving force for internal leakages, the applied pressure ratio r_{p}, is independent from the BVR, only the suction flow rate is normalised for this polynomial.

Subsequently, a polynomial for the compressor's isentropic efficiency is fitted based on a polynomial of the form

$$\begin{aligned} \eta_{\mathrm{is}} = {} & c_1 + c_2 x + c_3 y + c_4 x^2 + c_5 xy + c_6 y^2 + c_7 x^2 y + c_8 x y^2 + c_9 y^3 \\ & + c_{10} x^2 y^2 + c_{11} x y^3 + c_{12} y^4 \end{aligned} \tag{21}$$

using $\dot{V}_{\mathrm{suc,N}}$ and $r_{\mathrm{v,N}}$ for x and y respectively. The normalised applied volume ratio $r_{\mathrm{v,N}}$ is obtained from r_{v} by dividing it by the BVR analogously to Eq. (17).

In order to account for the rather small and approximately linear decline of η_{is} for larger pressure ratios (cf. Fig. 3(a)), the polynomial is blended over into a linearisation in the direction of $r_{\mathrm{v,N}}$ at a certain threshold value $r'_{\mathrm{v,N}}$.

This allows to accurately match the machine behaviour at small pressure ratios while avoiding unphysical behaviour due to the fourth-order nature of the polynomial at larger pressure ratios. Effectively, Eq. (21) is used to calculate η_{is} for all $r_{v,N} \leq r'_{v,N}$, while a linear function of the form

$$\eta_{is,lin} = \eta_{is}\left(\dot{V}_{suc,N}, r'_{v,N}\right) + \frac{d\eta_{is}}{dr_{v,N}}\left(\dot{V}_{suc,N}, r'_{v,N}\right) \cdot \left(r'_{v,N} - r_{v,N}\right) \tag{22}$$

is used for $r_{v,N} > r'_{v,N}$. Both Eqs. (21) and (22) return the isentropic efficiency as a percentage value. The final compressor model outputs (speed n, shaft power P_{sh} and specific outlet enthalpy h_{out}) are ultimately calculated based on the Eqs. (23) to (25):

$$n = \frac{\dot{V}_{suc}}{V_s \cdot \eta_{vol}} \tag{23}$$

$$h_{out} = \frac{\left(h_{out,is} - h_{in}\right) \cdot 100}{\eta_{is}} + h_{in} \tag{24}$$

$$P_{sh} = \dot{m} \cdot \left(h_{out} - h_{in}\right) \tag{25}$$

In these equations, h_{in} is the specific enthalpy at compressor inlet while $h_{out,is}$ is the specific enthalpy after isentropic compression. It is noted, that neglecting thermal losses in Eq. (25) inevitably leads to some inaccuracy. This will be discussed in more detail in Section 3.2.1 together with the validation results.

The different orders of the previously presented polynomials result from the model development strategy. The lowest possible order resulting in a good fit to the fitting data was chosen. The overarching goal behind this strategy was to reduce the number of polynomial coefficients while simultaneously reducing the risk of abrupt changes in the polynomial's behaviour outside the fitting range.

3. Results

3.1. Semi-empiric model results

This section presents the results of the fitting and validation procedure for the semi-empiric model described in Section 2.1. It further presents the operational data generated from the semi-empiric model and discusses noteworthy peculiarities of the model together with their implications on the further modelling approach.

3.1.1. Fitting and validation results

The compressor model is fitted and validated using catalogue data provided by the manufacturer Bitzer Kühlmaschinenbau GmbH (2023) [31] for the open-drive twin-screw compressor OSKA8591 using the refrigerant R134a as working fluid. Fig. 2 shows the fitting data distribution alongside the respective fitting and validation results. The machine exhibits a BVR of 3.1 and a swept volume of 3.075 L per shaft rotation. In order to provide a good database for the fitting and validation procedure, sixty-two data points are generated with a software tool provided by the manufacturer based on experimental data. For given evaporation and condensation temperatures (corresponding to the inlet and outlet pressure) and a specified compressor speed, the software calculates the necessary shaft power, the gas outlet temperature and the mass flow rate through the compressor. Thirty-one data points are created for compressor speeds of 2900 rpm and 3500 rpm each, with an inlet superheating of 10 °C.

Fig. 2(a) shows the corresponding temperatures in the data set used to fit and validate the semi-empiric model. A total number of twelve data points (six per speed level) are used to fit the model and determine its parameters. The data points used for fitting are evenly distributed among the operational range as indicated in Fig. 2(a). Using more data points for the fitting procedure only increases the required time without improving the model accuracy. The subfigures Fig. 2(b) and (c) show the validation of the fitting procedure in the form of parity plots for the shaft power and outlet temperature. Both parameters are predicted with high accuracy and show small values for the root mean square error (RMSE) calculated from the validation data points. Fig. 2(d) shows the relative deviation of the calculated compressor speed from the catalogue speed over the corresponding catalogue shaft power. Due to the limitation on only two catalogue speed levels, a parity plot like in subfigures (b) and (c) makes no sense for the compressor speed. It can be observed, that there is a linear trend in the compressor speed deviation with increasing shaft power. This is attributed to the model increasingly underestimating the compressor speed with rising outlet pressure. Looking at Eq. (7), this can be caused by two things. Either an underestimation of the internal leakage, resulting in a lower swept mass flow rate $\dot{m}_s$, or an underestimation of the suction heating, resulting in a lower specific volume v_3. Both leakage and heat transfer are calculated based on simplifying assumptions that cannot be independently validated with available data. For this reason, and since the compressor speed deviation is generally low, the linear trend in speed deviation is not further investigated and the model is deemed suitable for further use. The final model parameters determined in the fitting procedure are summarised in Table 1. To calculate the suction heat transfer coefficient according to Eq. (4), a nominal mass flow rate of 1 kg/s is used.

Table 1
Model parameters of the semi-empiric model used for data generation.

Parameter	A_{leak} [mm^2]	b_{hl} [W·K$^{-5/4}$]	$UA_{su,nom}$ [W/K]	$a_{tl,1}$ [−]	$a_{tl,2}$ [−]
Value	13.4394	2.9579	32.5083	0.2036	66.7533

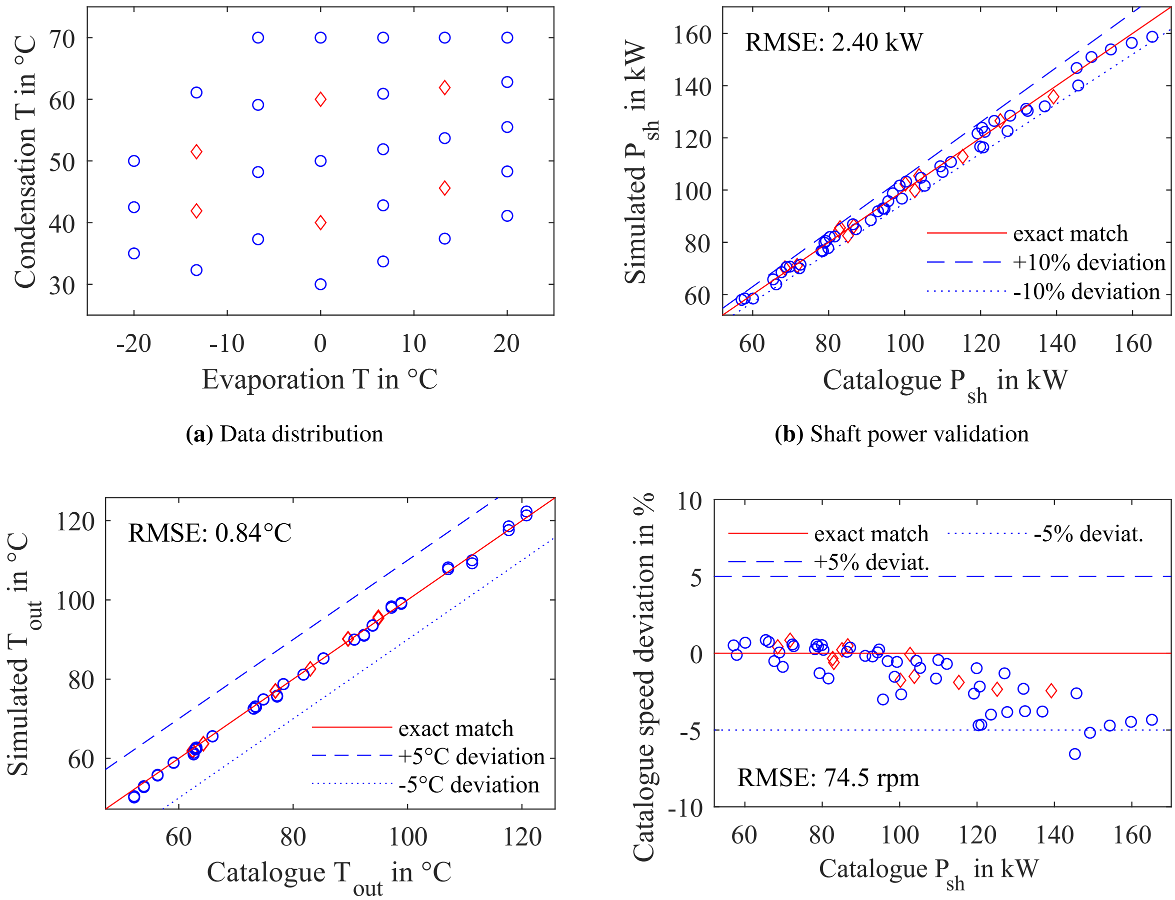
3.1.2. Semi-empiric model behaviour

Fig. 3 shows the isentropic and volumetric efficiencies calculated from the semi-empiric model over the operating range described in Section 2.1.2. Both efficiencies display the typical behaviour as it is reported for TSCs in literature. The isentropic efficiency drops sharply for small pressure ratios due to over-compression losses, while the decline in efficiency for larger pressure ratios (due to under-compression) is less pronounced. The location of the maximum efficiency with respect to the pressure ratio can be accounted to the machine's BVR of 3.1.

As expected, the volumetric efficiency drops significantly for larger pressure ratios since the driving force for internal leakages between the individual compression chambers increases. Moreover, an increasing flow rate (and hence increasing speed) leads to higher volumetric efficiency as the ratio between internal leakages and total conveyed fluid decreases. This is due to the effect that leakage is only proportional to the pressure differential while the total suction flow rate also depends on the compressor speed. The maximum isentropic efficiency obtained over the considered operational range is around 79 % which is very close to the maximum efficiency of the experimental data (around 78 %).

Looking at Fig. 3, it becomes apparent that the inlet pressure has a negligible effect on the volumetric efficiency, while there are distinct differences between the isentropic efficiencies for different inlet pressures. For inlet pressures of 1, 3 and 5 bar, average volumetric efficiencies of 89.47 %, 90.05 % and 92.90 % and average isentropic efficiencies of 65.24 %, 67.16 % and 72.59 % are obtained, respectively. The minor effect on the volumetric efficiency can be explained by Fig. 4(a), which shows the leakage mass flow rate (left y-axis) and the ratio between leakage and inlet mass flow rate (right y-axis) over the applied pressure ratio for different inlet pressures. As depicted, higher inlet pressures result in higher absolute leakage mass flow rates due to the generally higher pressure differential over the compressor, which is the driving force for internal leakage. The maximum applied pressure differential is 21 bar and 20 bar for inlet pressures of 3 bar and 5 bar, respectively, resulting in nearly the same maximum leakage mass flow rate. In contrast, the maximum differential is 7 bar for an inlet pressure of 1 bar, yielding significantly lower leakage.

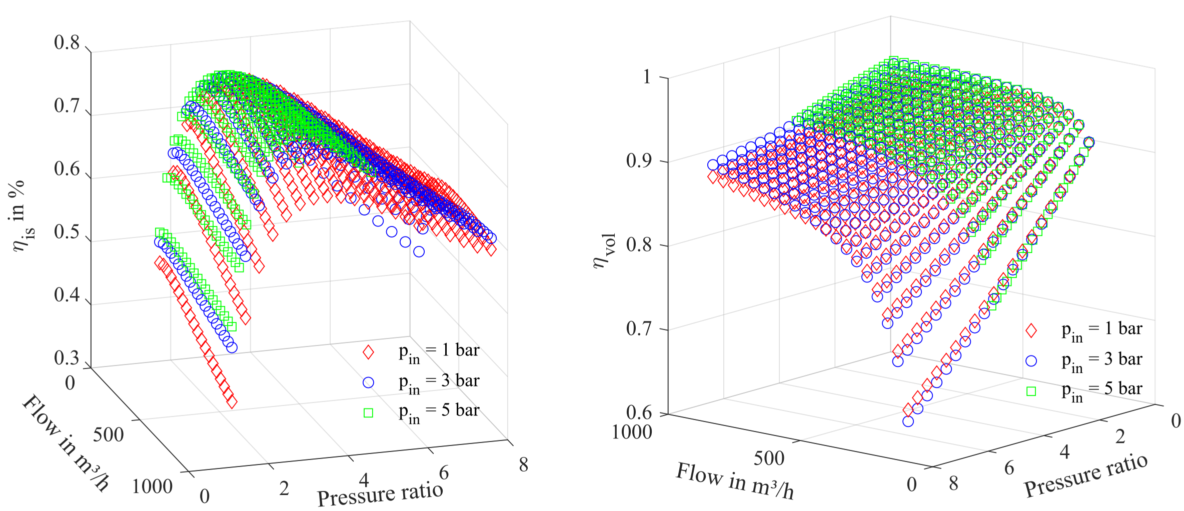
Since the inlet density of the fluid rises with increasing inlet pressures, a higher inlet mass flow rate is obtained at the same rotational speed. This leads to relatively minor differences in the ratio of internal



(a) Data distribution **(b)** Shaft power validation

(c) Outlet temperature validation **(d)** Compressor speed validation

Fig. 2. Fitting data distribution (a) as well as fitting results and validation (b-d) for the semi-empiric model (fitting data: red diamonds, validation data: blue circles).

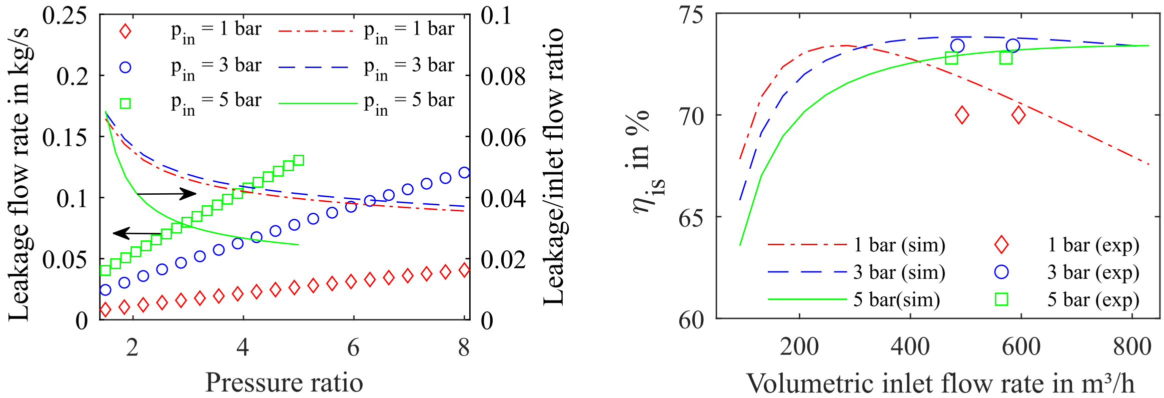


(a) Isentropic efficiency **(b)** Volumetric efficiency

Fig. 3. Efficiencies of the semi-empiric base model.

leakage and inlet mass flow rate for all observed inlet pressures (as depicted in Fig. 4(a)), explaining the small effect on the volumetric efficiency.

The differences in the isentropic efficiency, explicitly a strong decline with rising flow rates (and hence compressor speeds) for small inlet pressures, are not reflected in the underlying operational data. This is depicted in Fig. 4(b), which compares the simulated and experimental values of the isentropic efficiency over varying flow rates for a fixed pressure ratio of about 4.6. While the simulated efficiency values at inlet pressures of 3 bar and 5 bar are consistent with the experimental data from the manufacturer, a clear deviation is observed for 1 bar. This can be attributed to the calculation of the speed-dependent shaft power



(a) Leakage impact evaluation **(b)** Isentropic efficiency comparison

Fig. 4. Evaluation of the inlet pressure dependency of the leakage (a) and comparison of simulated and experimental efficiencies at a pressure ratio of approx. 4.6 (b).

losses defined in Eq. (11). In this equation, the oil viscosity is calculated based on the oil temperature using a function fitted to viscosity values from the oil's datasheet. The oil temperature is assumed to be the mean value between the heated suction temperature and the still-hot exhaust gas (equivalent to the fluid states 3 and 5 in Fig. 1). Naturally, the suction temperature is significantly lower for small inlet pressures leading to a higher oil viscosity and accordingly higher speed-dependent shaft power losses. This results in an over-proportional decline in shaft power for small inlet pressures leading to overall lower isentropic efficiency. A comparison with the manufacturer data also confirmed much higher oil inlet temperatures (compared to the refrigerant inlet temperatures) for small inlet pressures in the experimental setup. This cannot be replicated by the model in its current state.

For this reason, only the simulated operational data for inlet pressures of 3 bar and 5 bar are used in the fitting procedure of the polynomial model. With respect to the volumetric efficiency, this does not lead to any significant inaccuracies as described above.

3.2. Polynomial model results

This Section presents the results of the fitting procedure for the polynomial model described in Section 2.2. First, the polynomial model's ability to replicate the operating behaviour of compressors with varying swept volumes and BVRs is critically assessed. Then the influence of the working fluid on the prediction performance is discussed and a comparison with existing models from literature is conducted. Finally, the numerical cost of the polynomial and semi-empiric model is compared.

3.2.1. Fitting results and model accuracy

The polynomial parameters for $r_{v,p}$ and η_{vol} according to Eqs. (18) and (20) are obtained using MATLAB's curve fitting toolbox. The parameters of Eq. (21) and the optimal value for $r'_{v,N}$ are determined by a custom regression, minimising the mean square error of the polynomial model's isentropic efficiency (defined by Eqs. (21) and (22)) with respect to the efficiency data generated from the semi-empiric model. Table 2 provides the polynomial coefficients obtained for the respective parameters. The threshold value $r'_{v,N}$ is determined to be 1.3379.

The deviation between the polynomial model and the semi-empirical model is again visualised in the form of parity plots depicted in Fig. 5. The figure and the RMSE values provided in each subfigure indicate a very good fit between the polynomial model and the data generated from the base model.

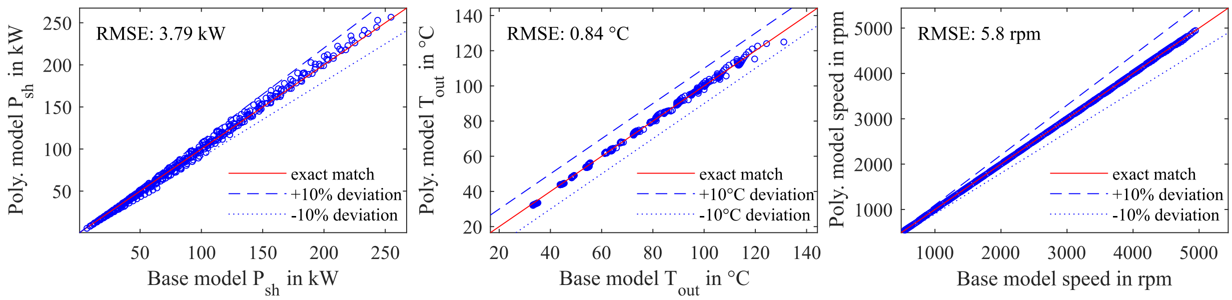
Since the parity plots in Fig. 5 only show the polynomial model deviation from the generated data set (cf. Section 2.1.2), another deviation analysis is made with the original operational data points from the manufacturer. This allows to evaluate potential amplifications of previous deviations between the manufacturer data and the semi-empiric model results which are not visible in Fig. 5. The parity plots in Fig. 6 provide the comparison of the manufacturer data and the polynomial model predictions. The plots confirm a good accuracy of the polynomial model also with respect to the initial operating data by the manufacturer. The limitation to fitting data generated with inlet pressures of 3 bar and 5 bar does not significantly reduce the polynomial model's prediction accuracy. It is noted, that the required shaft power is consistently slightly underestimated by the polynomial model (also c.f. Fig. 7(a)). This can likely be attributed to the neglect of thermal losses in the polynomial model. While the real machine loses some of the introduced shaft power in the form of heat losses to the environment, the polynomial model transfers the whole work input to the working fluid (either in the form of compression work or friction heat). Hence, the polynomial model requires a lower shaft power input to obtain the same outlet state. In extension to the RMSE values provided in Fig. 6, the model accuracy can be quantified by a mean absolute deviation from the manufacturer data of 4.02 %, 0.97 °C and 1.59 % for the shaft power, the outlet temperature and the compressor speed, respectively. The linear trend in compressor speed deviation in the Figs. 6(c) and 7(c) is a characteristic trait of the semi-empiric base model as described in Section 3.1.1. It is transferred to the polynomial model via the fitting procedure but does not reflect any characteristic behaviour of the polynomial model.

3.2.2. Model transferability testing

In order to evaluate the predictive capabilities of the polynomial model, it needs to be tested with machine sizes and BVR values different from the ones used during the model development. For this purpose, operational data for two smaller machines with a swept volume of 0.678 L is used. The respective machines are the models OSK5361-K (BVR of 3.1) and OSN5361-K (BVR of 4.7) from the same manufacturer Bitzer. The polynomial model's predictions are first directly compared to the operational data by the manufacturer, analogously to Fig. 6. In order to account for the size- and age-dependent maximum efficiency of the respective machine, an offset value is introduced into the polynomial model. It is the difference in maximum efficiency determined from the manufacturer data for the compared compressors within the observed operational area. The offset is added to the isentropic efficiency value yielded from Eq. (20) (or Eq. (22) when the linearisation is applied). Fig. 7 shows the validation for the OSK5361-K with an efficiency offset of -3 % using the same working fluid R134a. The parity plots for the OSK5361-K confirm very good model accuracy even at a significantly different swept volume when the BVR of the machine remains at the value used for the initial model development. A mean absolute deviation from the manufacturer data of 4.29 %, 0.88 °C and 1.38 % for

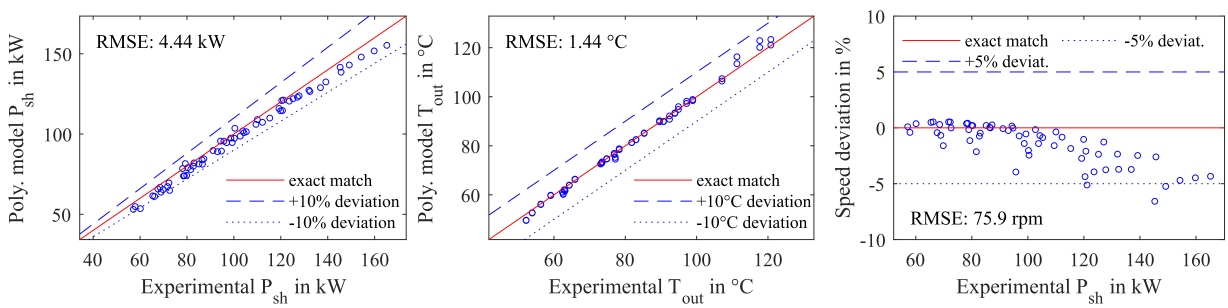
Table 2
Coefficients of the polynomial model.

Parameter	$r_{v,p}$	η_{vol}	η_{is}	Parameter	$r_{v,p}$	η_{vol}	η_{is}
c_1	0.870688	0.946384	−83.1921	c_{11}	−0.325559	−1.386582	−8.5495
c_2	0.006871	0.509457	−4.7660	c_{12}	0.275679	0.171104	−62.5854
c_3	0.200654	−0.083204	528.7869	c_{13}	−0.031426	0.007127	
c_4	−0.205568	−1.647231	2.8098	c_{14}	−0.002664	−0.000170	
c_5	0.204428	0.236403	−7.6153	c_{15}	0.086983	−0.000025	
c_6	−0.184384	0.001298	−629.7470	c_{16}	−0.072919	0.317757	
c_7	0.418683	2.247307	−4.9671	c_{17}	0.005582	−0.036833	
c_8	0.356840	−0.297059	30.6016	c_{18}	0.002952	−0.002011	
c_9	0.032637	−0.007651	320.7168	c_{19}		−0.000041	
c_{10}	0.030113	0.000305	−2.8907	c_{20}		0.000026	



(a) Shaft power deviation **(b)** Outlet temperature deviation **(c)** Compressor speed deviation

Fig. 5. Parity plots for comparison between base model and polynomial model.



(a) Shaft power deviation **(b)** Outlet temperature deviation **(c)** Compressor speed deviation

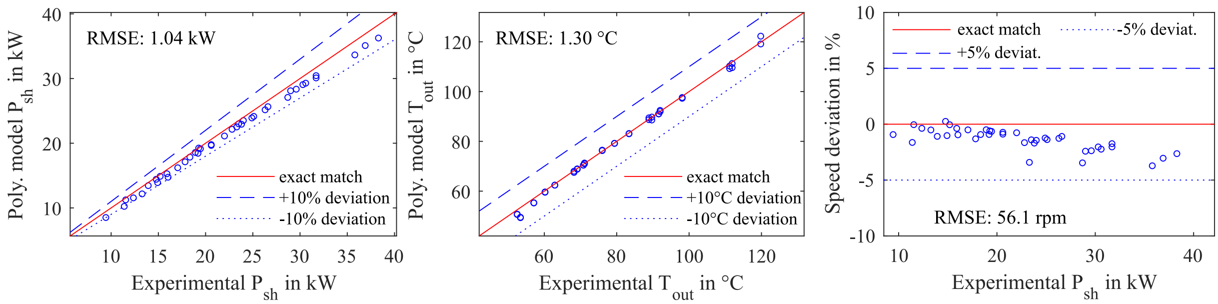
Fig. 6. Parity plots for comparison between manufacturer data and polynomial model.

the shaft power, the outlet temperature and the compressor speed can be observed, respectively.

Fig. 8 shows the parity plots for the model OSN5361-K with an efficiency offset of -7.2 % and the working fluid R407A, which is a mixture consisting of 40 % R134a, 40 % R125 (very similar to R134a) and 20 % R32. Unfortunately, no operational data with only R134a was available for this machine. Consequently, with R407A the most similar fluid with data available was chosen. An examination of the effect of different working fluids is provided in Section 3.2.3. It supports the assumption that the subsequently described deviations can mainly be attributed to the changing BVR.

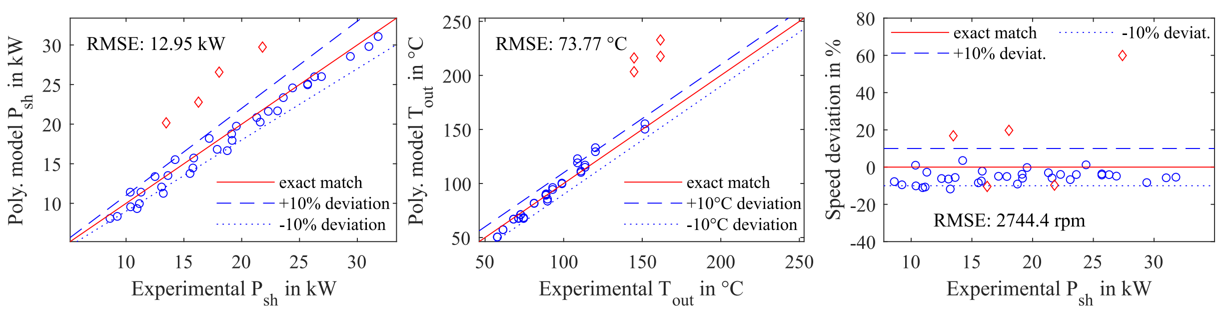
While the general predictive quality for a combination of differing swept volume and BVR is still quite good, larger deviations from the manufacturer data can be observed for very high pressure ratios. This is highlighted by the use of the red diamond markers in Fig. 8 for all operating points with an external pressure ratio larger than 14. For these pressure ratios, the polynomial model significantly underestimates the compressor efficiency, leading to higher shaft power input and higher outlet temperatures. It has to be mentioned that two operational points with external pressure ratios of around 22 are not depicted in Fig. 8 for the sake of readability. For such high pressure ratios, the linearisation of the isentropic efficiency results in negative efficiency values and accordingly negative shaft powers and outlet temperatures. Hence, to ensure physically meaningful results, the model should not be used with very large pressure ratios, especially, when higher BVR values are used.

As the smallest external pressure ratio in the manufacturer data for the OSN5361-K is around 4, the polynomial model is also compared to operational data generated from the semi-empiric model for the OSN5361-K, including smaller pressure ratios down to 1.5. The semi-empiric model is fitted according to the procedure described in Section 3.1.1 and subsequently used to create the same operational data set described in Section 2.1.2. For the sake of simplicity, only an inlet pressure of 3 bar is considered. When compared to the data from the semi-empiric model, larger deviations of the polynomial model can be observed for small pressure ratios. This is highlighted by the use of the red diamond markers for all operating points with an external pressure ratio smaller than 2.9 in Fig. 9. For these pressure ratios, the polynomial model significantly underestimates the compressor efficiency, leading to higher shaft power input and higher outlet temperatures. The deviations for the OSN5361-K can be explained by looking at the operational behaviour of two machines with the same swept volume but different BVR values. Fig. 10(a) shows the isentropic efficiency predicted by the semi-empiric models for the machines OSK5361-K (BVR of 3.1) and OSN5361-K (BVR of 4.7) over the normalised pressure ratio. The respective model parameters are summarised in Table 3.



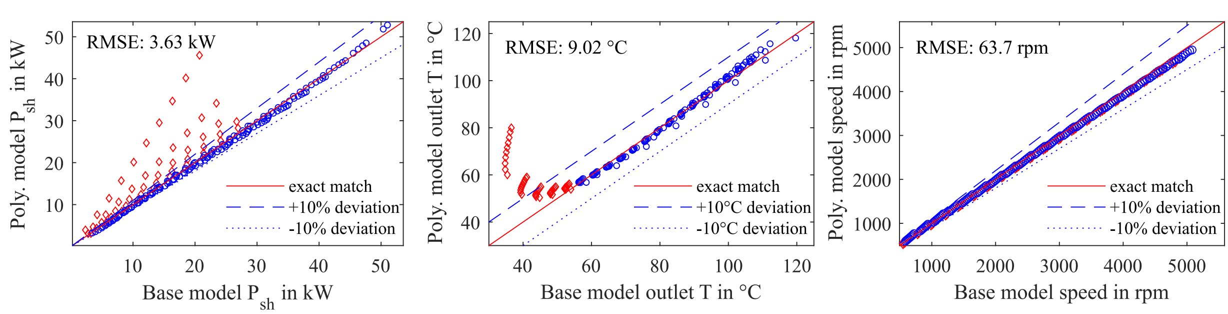
(a) Shaft power deviation **(b)** Outlet temperature deviation **(c)** Compressor speed deviation

Fig. 7. Model transferability validation for the OSK5361-K with smaller V_s and same BVR.



(a) Shaft power deviation **(b)** Outlet temperature deviation **(c)** Compressor speed deviation

Fig. 8. Model transferability validation for the OSN5361-K with different V_s and BVR (red diamonds: r_p > 14, blue circles: r_p < 14).



(a) Shaft power deviation **(b)** Outlet temperature deviation **(c)** Compressor speed deviation

Fig. 9. Model transferability validation for the OSN5361-K with different V_s and BVR (red diamonds: r_p < 2.9, blue circles: r_p > 2.9).

Table 3
Model parameters of the semi-empiric models for the compressors OSK5361-K and OSN5361-K.

Compressor	A_{leak} [mm^2]	b_{hl} [W·K$^{-5/4}$]	$UA_{su,nom}$ [W/K]	$a_{tl,1}$ [−]	$a_{tl,2}$ [−]
OSK5361-K	2.9271	0.7382	32.879	0.2566	33.88
OSN5361-K	2.0932	4.4612	202.533	0.2794	178.04

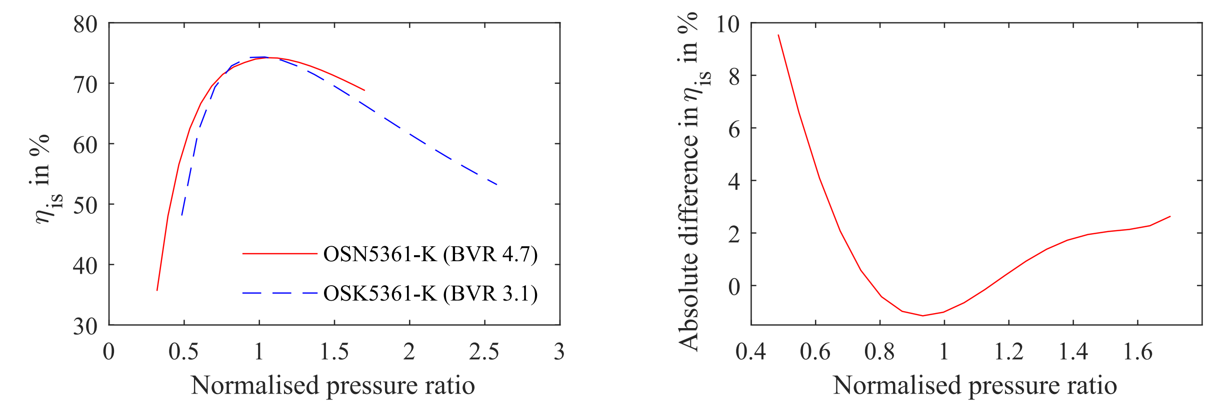
The values for all flow rates at each normalised pressure ratio were averaged to obtain a 2D graph. Moreover, an efficiency offset of +4.2 % was added to the efficiency of the OSN5361-K to obtain the same maximum efficiency and enable a fair comparison.

The figure shows that the efficiency begins to drop sharper for both negative and positive deviations from the ideal pressure ratio for the OSK5361-K, which shows very similar behaviour to the OSKA8591 used in the polynomial model development. This leads to a significant difference in efficiency between the machines with different BVRs. Looking at the deviations in Figs. 8 and 9 it becomes clear that the effect is more pronounced for smaller pressure ratios. This is also reflected in Fig. 10(b), which shows the difference in isentropic efficiency for the range of overlapping curves in Fig. 10(a). Here, the stronger increase in efficiency deviation for smaller normalised pressure ratios compared to larger ones becomes apparent. It can be concluded that major differences in BVR are a limitation of the proposed model in its current form, especially when the external pressure ratio deviates from the compressor's BVR values significantly.

It is noted that the model transferability testing for machines with higher swept volumes (and hence higher power scales) cannot be included in this work due to the unavailability of suitable validation data. However, the basic construction and, accordingly, the characteristic efficiency curves do not change in a relevant way for larger TSCs. There is no general trend and there is consequently no reason to expect major prediction inaccuracies for larger TSCs. An exception to this general validity can be appointed to TSCs with a variable BVR, which generally show efficiency curves that are stretched in the direction of the pressure ratio. Here, some deviation of the polynomial model can be expected.

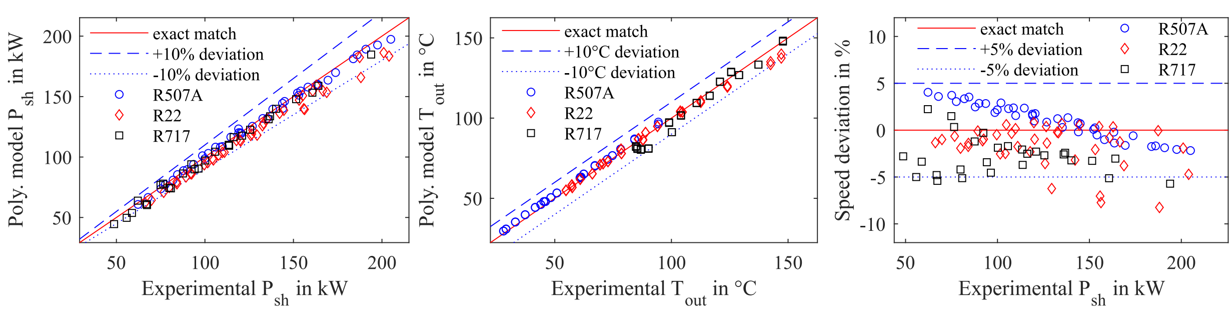
3.2.3. Validation for different fluids

The thermo-physical properties of the working fluid are undoubtedly a major influence on the performance of a compressor. To evaluate



(a) (b)

Fig. 10. Mean isentropic efficiency predicted by the semi-empiric models (a) and difference in predicted efficiency (b) over the normalised external pressure ratio of two machines with the same swept volume but different BVR.



(a) Shaft power deviation **(b)** Outlet temperature deviation **(c)** Compressor speed deviation

Fig. 11. Model transferability validation for different fluids, i.e. R507A, R22 and Ammonia (R717), used in the compressor OSK8591-K.

the model transferability for different working fluids, a comparison of polynomial model prediction and experimental compressor performance is conducted for three different working fluids. The specific fluids are the refrigerants R507A (consisting of 50 % R125 and 50 % R143a), R22 and R717 (ammonia). The compressor used to obtain the experimental results is a Bitzer OSK8591 (i.e. the same machine used in the initial model development). Fig. 11 shows the comparison results. The parity plots for shaft power and outlet temperature in Fig. 11(a) and (b) show good agreement between polynomial model predictions and experimental values. The deviations are mostly in the same range as for the experimental model validations in Fig. 6 where the working fluid used in the fitting procedure (i.e. R134a) is compared. Also the compressor speed deviation depicted in Fig. 11(c) shows only small deviations for the different observed fluids, indicating a good prediction accuracy of the volumetric efficiency polynomial for different fluids. Naturally, this does not conclude that the polynomial model can be used independently from any fluid properties. The model results should be assessed with caution for fluids with strong deviations in thermophysical properties (compared to the fluids studied in this work). Yet, the validation for different fluids indicates that the goal of a generalised model is achieved and no inherent major inaccuracies are introduced by using different fluids.

3.2.4. Comparison to literature models

To highlight the benefits of the generalised polynomial model developed in this work, a comparison with two models from literature is presented subsequently. In explicit, the empirical models of Kosmadakis and Neofytou (2022) [11] and Liu et al. (2012) [1] are used for the comparison. The former model uses the applied volume ratio and the volumetric inlet flow rate to calculate the isentropic efficiency and was fitted for a compressor with a BVR of 3 and a swept volume of 2.972 L. The latter model yields the isentropic efficiency based on the applied outlet pressure, the BVR and the compression's polytropic index. It was fitted on several compressors with BVR values between 2.2 and 3.3. Fig. 12 shows the model predictions of the semi-empirical model described in Section 2.1.1 (indicated by "S.E. Model" in the Figures), the novel polynomial model and both aforementioned models from literature. Both models from literature show a clear dependency on the applied pressure ratio in Fig. 12(a). Due to the similar BVRs of all compressor models, the maximum efficiency is yielded at similar pressure ratios. However, the absolute values differ significantly for the models from literature. While the model of Liu et al. strongly overestimates the isentropic efficiency, the model of Kosmadakis and Neofytou shows the opposite tendency. This could, in theory, be improved by adding an efficiency offset to the literature model predictions similar to the procedure described for the smaller compressors in Section 3.2.2. The comparably less pronounced effect of over- and under-compression for the model of Kosmadakis and Neofytou indicates a high probability for flawed model predictions at small and large pressure ratios. The dependency of the isentropic efficiency on the volumetric inlet flow rate depicted in Fig. 12(b) highlights a clear advantage of the novel polynomial model. While the model of Liu et al. neglects the influence of the flow rate (and hence the compressor speed) completely, the model of Kosmadakis and Neofytou only yields physically meaningful results within a very narrow range.

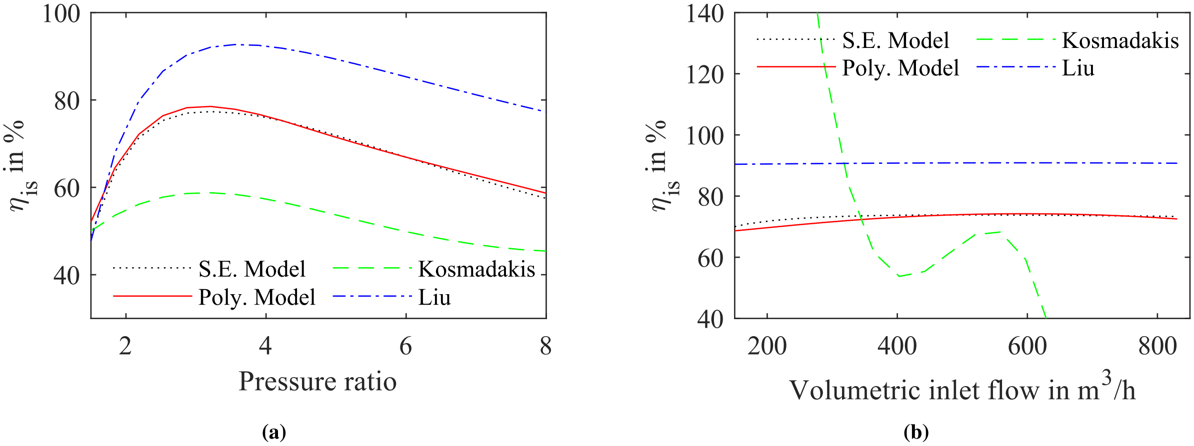


Fig. 12. Isentropic efficiency predicted by different compressor models over a varied pressure ratio at a volumetric inlet flow rate of 442 m^3/h (a) and over a varied volumetric inlet flow rate at a pressure ratio of 4.56 (b).

This is most likely caused by a narrow range of flow rates in their fitting procedure and the previously discussed behaviour of higher-degree polynomials outside the fitting range. The novel polynomial model is able to avoid this issue by normalising the inlet flow rate. Another major advantage of the model presented in this work is its explicit formulation. Both literature models rely on input parameters (i.e. the volume ratio for Kosmadakis and Neofytou and the polytropic index for Liu et al.) that are dependent on the compressor outlet temperature and hence not known a priori. While the authors do not account for this in their respective publications, it requires an iterative solver, which significantly increases the computational cost of their models.

3.2.5. Numerical cost comparison

A gain in computational speed is one major motivation in the previously described model development. Hence, the computational cost of the semi-empiric base model described in Section 2.1.1 and the polynomial model is subsequently compared. For comparison, the data set generation procedure described in Section 2.1.2 was repeated with a 20 × 20 grid, resulting in a total of 400 operational points calculated by the semi-empiric model. The same input data was fed to the polynomial model which also calculated the 400 operational points. In order to account for the varying load on the host machine's CPU caused by the operating system, this procedure was repeated six times. The results are given in Table 4. It can be seen that the required computational times differ depending on the momentary CPU load. However, the mean ratio between the required computation times t is, on average, 375.2, meaning that the polynomial model only requires 0.26 % of the semi-empiric model's solving time. This massive reduction in computational cost for the polynomial model has the potential to drastically reduce the computational time of complex cycle and system simulations, such as annual simulations with cycle optimisation.

Table 4
Computational time requirements of the semi-empiric and polynomial compressor models for 400 operational points and ratio of both.

$t_{\text{semi-empirical}}$ [s]	333.1	273.1	268.2	268.4	272.5	274.8
$t_{\text{polynomial}}$ [s]	0.8345	0.7148	0.7776	0.6964	0.7886	0.6974
Ratio [-]	399.2	382.1	344.9	385.3	345.6	394.0

4. Conclusion

This work presents a novel polynomial model for twin-screw compressors yielding mostly accurate machine performance predictions independent from the specific machine size and built-in volume ratio. The model is developed in two steps. First, an accurate semi-empiric model from literature is slightly adapted and used to generate performance data for a large operational field. In the next step, a polynomial model with a linearisation for high pressure ratios is fitted to this data. The linearisation allows to accurately simulate the characteristic machine behaviour at small and medium pressure ratios with a fourth-order polynomial while avoiding unphysical behaviour caused by the mathematical properties of a fourth-order polynomial at larger pressure ratios. The model uses the external pressure ratio and volumetric compressor suction flow rate to calculate isentropic efficiency and compressor speed. Both input parameters are normalised with a reference flow rate (calculated from the swept volume) and the BVR, respectively. This results in a generalised model of low numerical cost, which can be used for explorative studies independent of the specific machine size and BVR. When used to predict the performance of machines with similar BVR (3.1 for model development) but different sizes, the model displays very good predictive accuracy. However, when there is a difference in size and BVR, the prediction accuracy is still reasonable but significantly declines for small or very large pressure ratios. For strong deviations in BVR (compared to the fitting value of 3.1), users of the polynomial model should be careful when the normalised pressure ratio is below the value of 0.6 or exceeds a value of 3.

Nevertheless, the presented methodology for accurate and time-efficient models might be of high interest to the heat pump and general compressor community enabling the integration of more precise models in computationally intensive simulation and optimisation routines. This is further supported by the validation of the polynomial model for different fluids, which confirms good prediction accuracy for different fluids. The comparison of the computational cost between the semi-empiric reference model and the polynomial model shows a drastic reduction in required computational time. The polynomial model, on average, only requires 0.26 % of the time the semi-empiric model needs to be solved.

In the future, the proposed model could be transferred to similar volumetric compressors such as scroll machines, since they share key working principles and loss mechanisms of TSCs. It is also noted that the described modelling method can be adapted to obtain a polynomial model explicitly in compressor speed (instead of flow rate) and pressure ratio if the intended application requires. Moreover, the model can be adapted for twin-screw expanders (TSEs), which show a similar construction and hence highly similar efficiency curves as TSCs. This can make the polynomial model's advantages available for extensive simulations of processes where part load of TSEs is common and important (such as Organic Rankine Cycles).

Nomenclature

Abbreviations

BVR	built-in volume ratio
ORC	Organic Rankine Cycle
RMSE	root mean square error
TSC	twin-screw compressor

Subscripts

amb	ambient
ex	exhaust
hl	heat loss
int	internal
is	isentropic
leak	leakages
lin	linearised
N	normalised
nom	nominal
p	with respect to pressure
ref	reference
s	swept
sh	shaft
su	after inlet heating
suc	suction at compressor inlet
tl	torque loss
v	with respect to volume
vol	volumetric
w	wall

Symbols

a	dimensionless model parameter, –
A	area, mm^2
c_p	isobaric heat capacity, J/(kg K)
$\dot{m}$	mass flow, kg/s
h	specific enthalpy, J/kg
n	rotational speed, rpm or Hz
Nu	Nusselt number, –
p	pressure, Pa
P	power, W
Pr	Prandtl number, –
$\dot{Q}$	heat flow, W
r	ratio, –
s	specific entropy, J/(kg K)
Re	Reynolds number, –
t	time, s
T	temperature, K or °C
UA	lumped heat transfer factor, W/K
v	specific volume, m^3/kg
V	volume, m^3
$\dot{V}$	volumetric flowrate, m^3/s
γ	isentropic exponent, –
η	isentropic efficiency, – or %
μ	dynamic viscosity, Pa s

CRediT authorship contribution statement

Florian Kaufmann: Methodology, Software, Data curation, Investigation, Visualisation, Writing – original draft. **Ludwig Irrgang:** Writing – review & editing. **Christopher Schifflechner:** Writing – review & editing, Supervision, Project administration. **Hartmut Spliethoff:** Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

[1] J. Liu, Q. Li, F. Wang, L. Zhou, A new model of screw compressor for refrigeration system simulation, Int. J. Refrig. 35 (4) (2012) 861–870, http://dx.doi.org/10.1016/j.ijrefrig.2012.01.016, URL: http://www.sciencedirect.com/science/article/pii/S014070071200028X.

[2] S. Eyerer, F. Dawo, M. Altnöder, R. Windhager, A. Niederdränk, H. Esch, S. Ausfelder, R. Konrad, M. Höger, C. Wieland, H. Spliethoff, Design and first operation of an advanced ORC-CHP architecture, in: Proceedings of the 5th International Seminar on ORC Power Systems, 2019, URL: https://www.orc2019.com/online/proceedings/display_manuscript/27.htm.

[3] H. Öhman, Low temperature difference power systems and implications of multiphase screw expanders in Organic Rankine Cycles, in: TRITA-REFR, vol. 15/02, KTH Royal Institute of Technology, Stockholm, 2016, URL: http://kth.diva-portal.org/smash/record.jsf?pid=diva2%3A934157&dswid=4728.

[4] F. Dawo, J. Buhr, C. Schifflechner, C. Wieland, H. Spliethoff, Experimental assessment of an Organic Rankine Cycle with a partially evaporated working fluid, Appl. Therm. Eng. 221 (2023) 119858, http://dx.doi.org/10.1016/j.applthermaleng.2022.119858, URL: https://www.sciencedirect.com/science/article/pii/S1359431122017884.

[5] M. Weitzer, S. Kolb, D. Müller, J. Karl, Reversible heat pump-ORC pilot plant - experimental results and fluid charge optimization, in: Proceedings of the 7th International Seminar on ORC Power Systems, Universidad de Sevilla, 2023, URL: https://alojaservicios.us.es/difuseditorial/Extra_content/orc_ps_24/orc_editorial_072.pdf.

[6] P. Byrne, R. Ghoubali, J. Miriel, Scroll compressor modelling for heat pumps using hydrocarbons as refrigerants, Int. J. Refrig. 41 (2014) 1–13, http://dx.doi.org/10.1016/j.ijrefrig.2013.06.003.

[7] C. Cuevas, J. Lebrun, V. Lemort, E. Winandy, Characterization of a scroll compressor under extended operating conditions, Appl. Therm. Eng. 30 (6) (2010) 605–615, http://dx.doi.org/10.1016/j.applthermaleng.2009.11.005, URL: http://www.sciencedirect.com/science/article/pii/S1359431109003330.

[8] M.W. Browne, P.K. Bansal, Transient simulation of vapour-compression packaged liquid chillers, Int. J. Refrig. 25 (5) (2002) 597–610, http://dx.doi.org/10.1016/S0140-7007(01)00060-3.

[9] E. Navarro-Peris, J.M. Corberán, L. Falco, I.O. Martínez-Galván, New non-dimensional performance parameters for the characterization of refrigeration compressors, Int. J. Refrig. 36 (7) (2013) 1951–1964, http://dx.doi.org/10.1016/j.ijrefrig.2013.07.007.

[10] D.R. Markham, J.M. Cutbirth, Performance of a custom asymmetric screw compressor, in: SAE Technical Paper Series, SAE International400 Commonwealth Drive, Warrendale, PA, United States, 2016, http://dx.doi.org/10.4271/2016-01-1996.

[11] P. Neofytou, Reversible high-temperature heat pump/ORC for waste heat recovery in various ships: A techno-economic assessment, Energy 256 (2022) 124634, http://dx.doi.org/10.1016/j.energy.2022.124634, URL: https://www.sciencedirect.com/science/article/pii/S0360544222015377.

[12] A. Giuffrida, A semi-empirical method for assessing the performance of an open-drive screw refrigeration compressor, Appl. Therm. Eng. 93 (2016) 813–823, http://dx.doi.org/10.1016/j.applthermaleng.2015.10.023, URL: http://www.sciencedirect.com/science/article/pii/S1359431115010704.

[13] Z. Zhao, Z. Xing, F. Hou, Y. Tian, S. Jiang, Theoretical and experimental investigation of a novel high temperature heat pump system for recovering heat from refrigeration system, Appl. Therm. Eng. 107 (2016) 758–767, http://dx.doi.org/10.1016/j.applthermaleng.2016.07.047.

[14] N. Salts, D. Ziviani, E.A. Groll, Application of a generalized compressor modeling framework for simulating an oil-injected twin-screw compressor, IOP Conf. Ser.: Mater. Sci. Eng. 604 (2019) 012009, http://dx.doi.org/10.1088/1757-899X/604/1/012009.

[15] K. Hanjalić, N. Stošić, Development and optimization of screw machines with a simulation model—Part II: Thermodynamic performance simulation and design optimization, J. Fluids Eng. 119 (3) (1997) 664–670, http://dx.doi.org/10.1115/1.2819296.

[16] W.S. Lee, R.H. Ma, S.L. Chen, W.F. Wu, H.W. Hsia, Numerical simulation and performance analysis of twin screw air compressors, Int. J. Rotating Mach. 7 (1) (2001) 65–78, http://dx.doi.org/10.1155/S1023621X01000069.

[17] N. Seshaiah, S.K. Ghosh, R.K. Sahoo, S.K. Sarangi, Mathematical modeling of the working cycle of oil injected rotary twin screw compressor, Appl. Therm. Eng. 27 (1) (2007) 145–155, http://dx.doi.org/10.1016/j.applthermaleng.2006.05.007.

[18] D. Buckney, A. Kovacevic, N. Stošić, Experimental validation of a geometry model for twin screw machines, in: 8th International Conference on Compressors and their Systems, Woodhead Publishing, Cambridge, England, 2013, pp. 237–246, http://dx.doi.org/10.1533/9781782421702.4.237.

[19] M. Chamoun, R. Rulliere, P. Haberschill, J.-L. Peureux, Modelica-based modeling and simulation of a twin screw compressor for heat pump applications, Appl. Therm. Eng. 58 (1–2) (2013) 479–489, http://dx.doi.org/10.1016/j.applthermaleng.2013.04.020.

[20] Y.R. Wu, V.T. Tran, Dynamic response prediction of a twin-screw compressor with gas-induced cyclic loads based on multi-body dynamics, Int. J. Refrig. 65 (2016) 111–128, http://dx.doi.org/10.1016/j.ijrefrig.2015.12.002.

[21] S. Kennedy, M. Wilson, S. Rane, Combined numerical and analytical analysis of an oil-free twin screw compressor, IOP Conf. Ser.: Mater. Sci. Eng. 232 (2017) 012080, http://dx.doi.org/10.1088/1757-899X/232/1/012080.

[22] R. Andres, J. Hesse, F. Hetze, D. Low, Cfd simulation of a two stage twin screw compressor including leakage flows and comparison with experimental data, IOP Conf. Ser.: Mater. Sci. Eng. 425 (2018) 012018, http://dx.doi.org/10.1088/1757-899X/425/1/012018.

[23] N. Basha, A. Kovacevic, S. Rane, Analysis of oil-injected twin-screw compressor with multiphase flow models, Designs 3 (4) (2019) 54, http://dx.doi.org/10.3390/designs3040054.

[24] S.S. Byeon, J.Y. Lee, Y.J. Kim, Numerical analysis on the flow characteristics of oil-injected screw air compressor, in: 2014 ISFMFE - 6th International Symposium on Fluid Machinery and Fluid Engineering, IEEE, Piscataway, NJ, 2014, p. 045 (6 .), http://dx.doi.org/10.1049/cp.2014.1172.

[25] A. Kovacevic, S. Rane, N. Stošić, Modelling of twin-screw machines by use of CFD, in: T.H. Phung, I.A. Sultan (Eds.), Positive Displacement Machines: Modern Design Innovations and Tools, Academic Press, 2020, pp. 423–441, http://dx.doi.org/10.1016/B978-0-12-816998-8.00015-7.

[26] The MathWorks Inc., MATLAB, Natick, Massachusetts, United States, 2021.

[27] E.W. Lemmon, I.H. Bell, M.L. Huber, M.O. McLinden, NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties-REFPROP: Standard Reference Data Program, National Institute of Standards and Technology, Gaithersburg, United States, 2018.

[28] Bitzer Kühlmaschinenbau GmbH, Open Drive Screw Compressors: OS.(A)53 // OS.(A)74 Series, Sindelfingen, 2014.

[29] V. Lemort, S. Quoilin, C. Cuevas, J. Lebrun, Testing and modeling a scroll expander integrated into an organic rankine cycle, Appl. Therm. Eng. 29 (14–15) (2009) 3094–3102, http://dx.doi.org/10.1016/j.applthermaleng.2009.04.013.

[30] F.W. Dittus, L. Boelter, Heat transfer in automobile radiators of the tubular type, Int. Commun. Heat Mass Transfer 12 (1) (1985) 3–22, http://dx.doi.org/10.1016/0735-1933(85)90003-X.

[31] Bitzer Kühlmaschinenbau GmbH, BITZER software, 2023, URL: https://www.bitzer.de/websoftware/.